

Spatial Gene Expression from Histology

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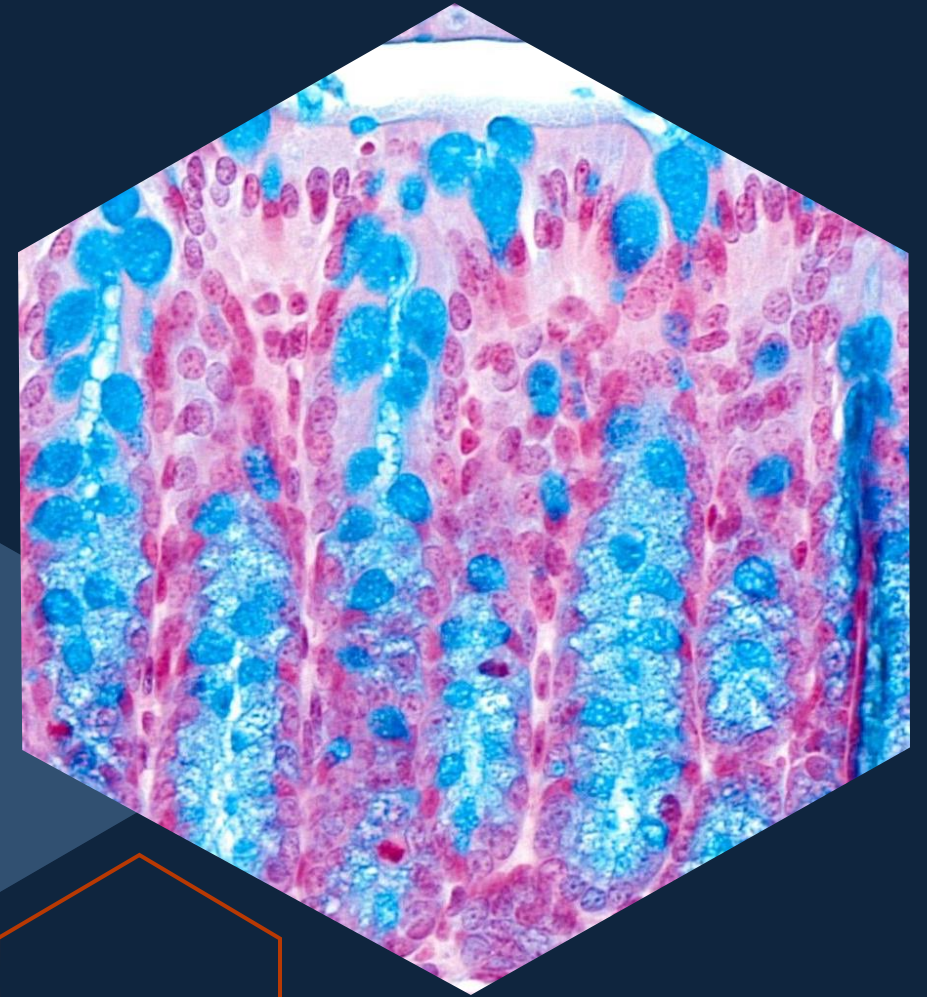


Agenda

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- Background
 - Objective overview
 - Potential Pathways
 - Resource management

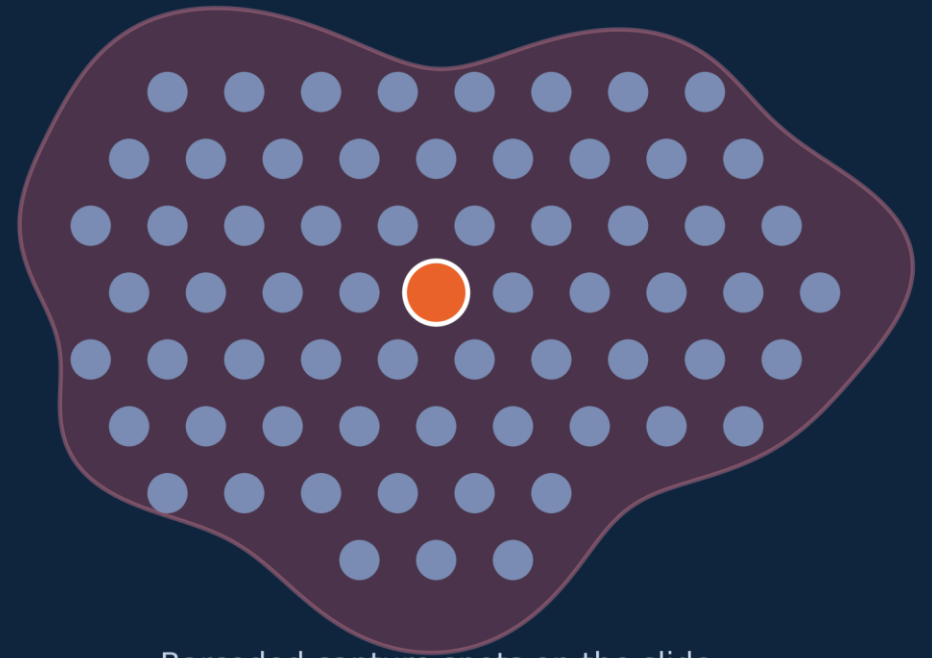
Current interpretation.....

Backgrounds



The problem

- Input: an H&E image patch
- Output: Gene expression at that spot
- Question: Does the picture predict the numbers, or just its neighbors?

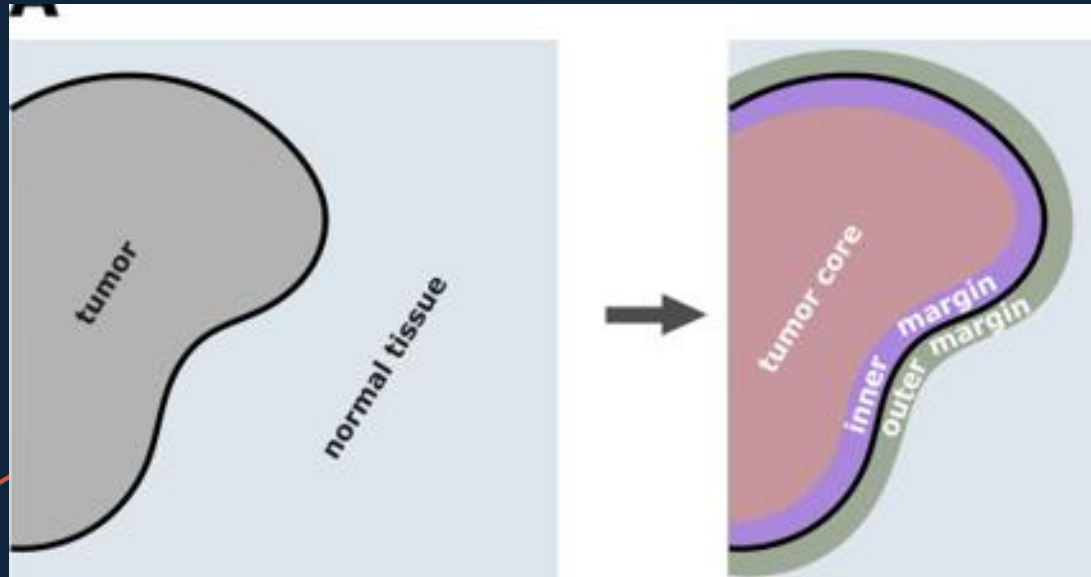


Barcoded capture spots on the slide



Expression values at that spot

Why per-location, not per-slide



Kather et al., eLife 2018, Fig. 1A (CC BY 4.0)

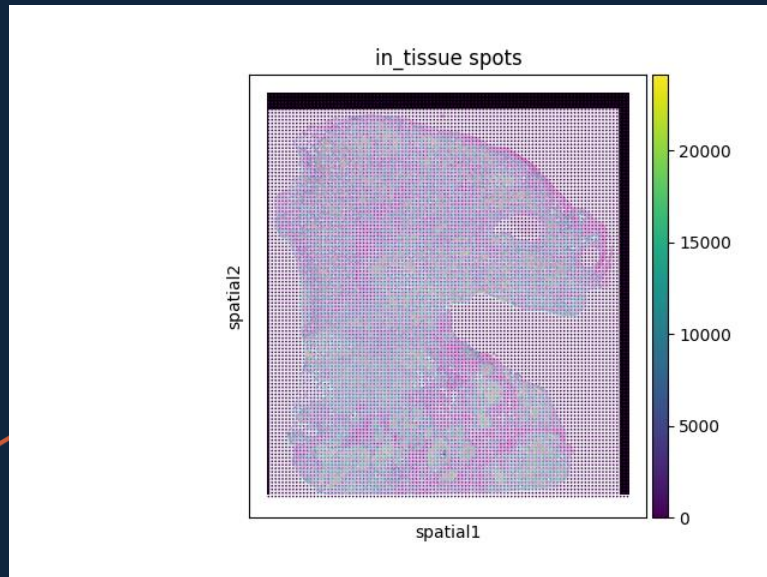
Tumor core and margin are different tissue, millimeters apart.

One number for the whole sample averages them together.

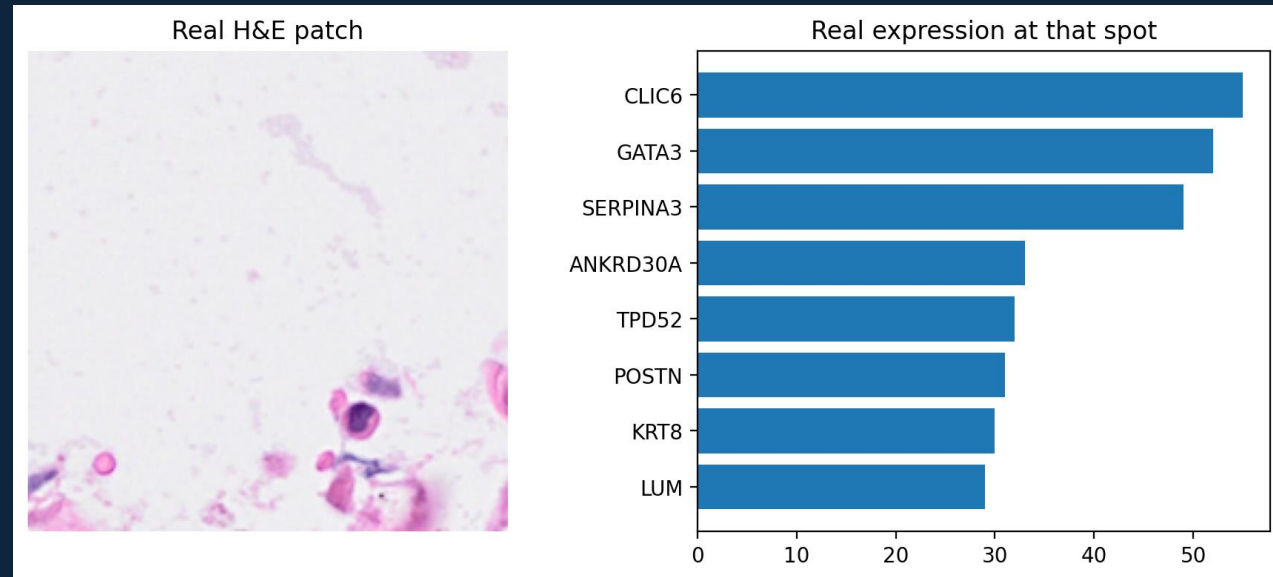
That is why the target has to be a value per spot, not per slide.



What the data looks like



Every capture spot, one real slide



One spot: real patch, real measured expression



Sample TENX95 — breast tissue, *Homo sapiens*, Xenium. Pulled directly from our HEST-1k cohort (Jaume et al., NeurIPS 2024, CC BY-NC-SA 4.0)

Dataset and target output

Dataset

HEST-1k: 1,276 paired H&E + spatial transcriptomics samples

Each sample: one whole-slide image and a grid of capture spots with measured expression

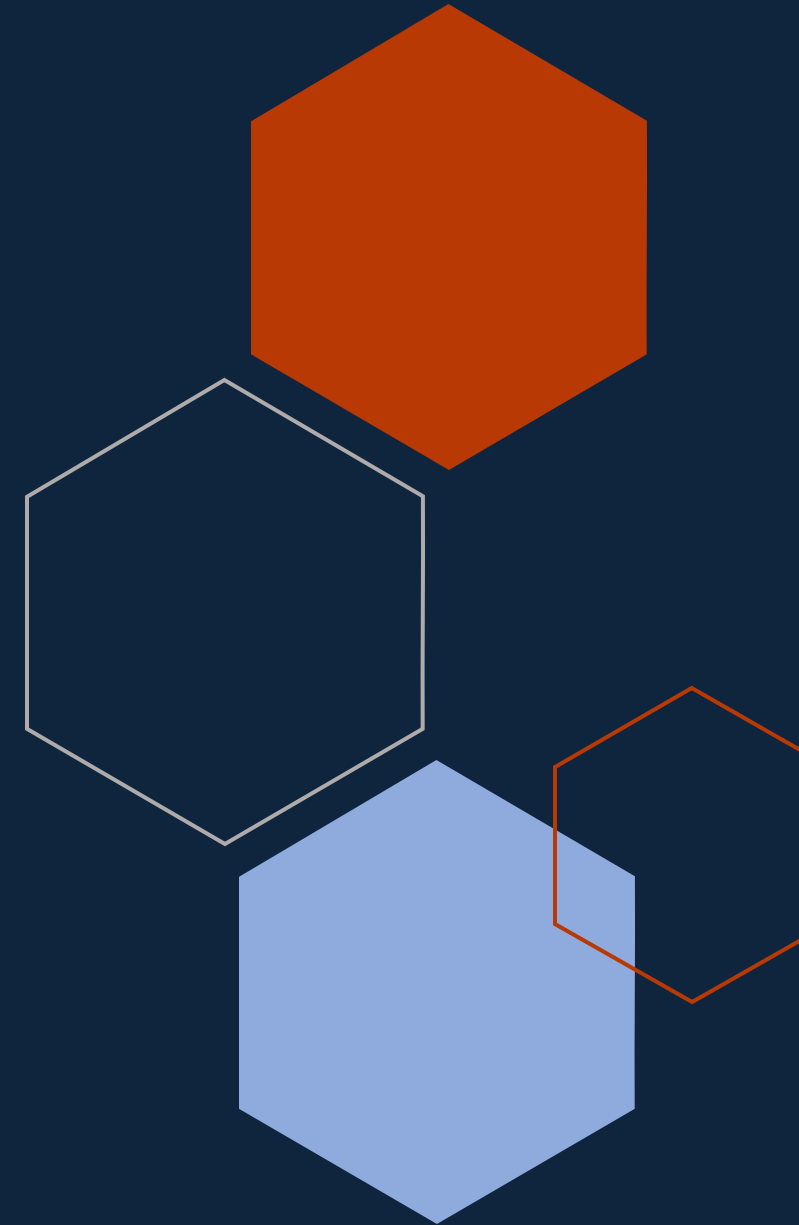
We use a selected subset, split by patient. Not by spot

Target output

For each spot: expression values for a selected set of genes

Model input: the H&E patch under that spot

Scored by correlation between predicted and true expression, gene by gene



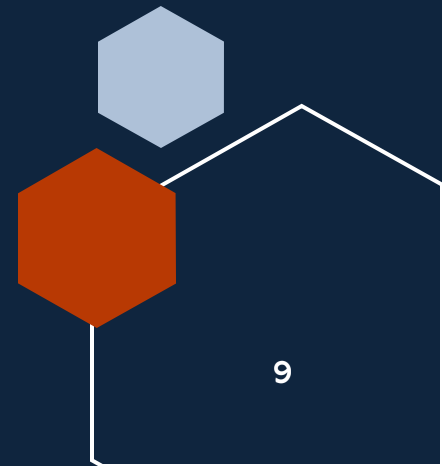
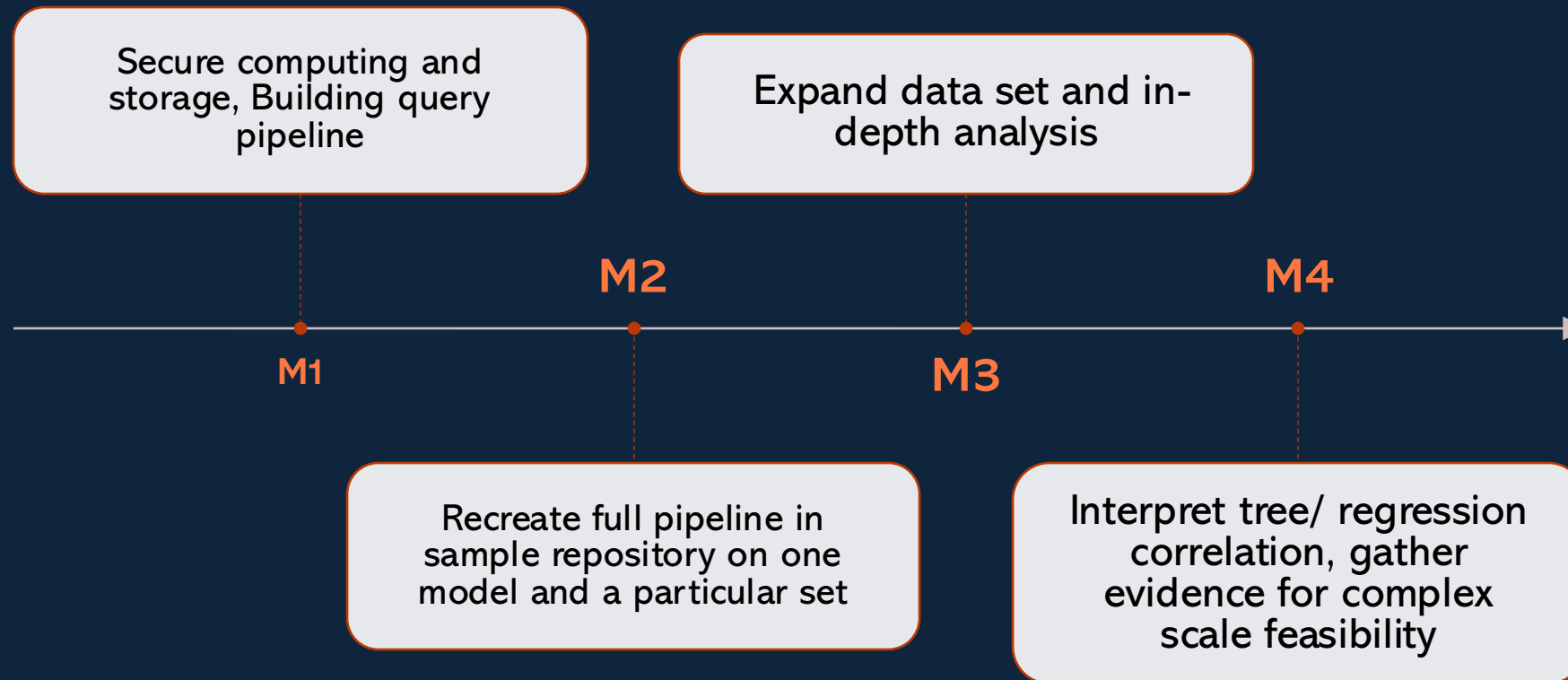


Objectives

1. Overall goal:
Predict gene expression with histology stains
2. Restraints:
Consider only a few genes
Only a subset of the data
3. Evaluation:
Compare the difference between output and ground truth (in dataset)

First stage Timeline

- A chronological list of priorities
- Exact time will be added later

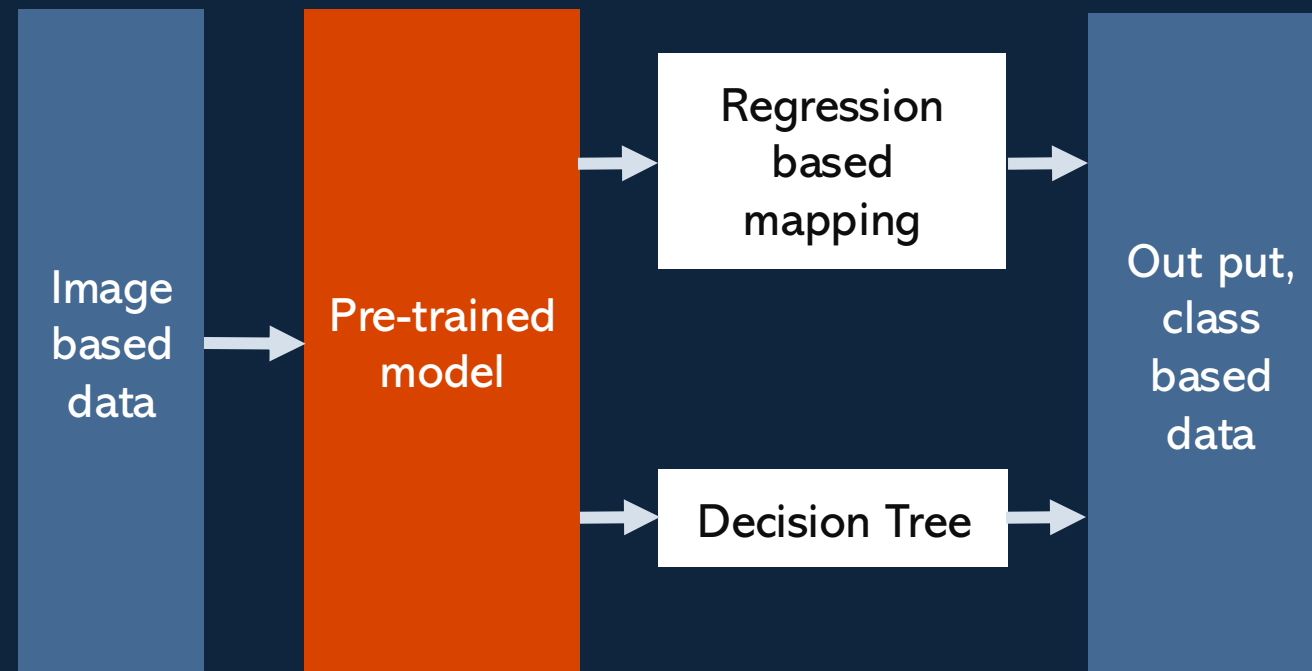


Potential Pathways

- Utilizing pretrained, large models for embedding
- Use shallow and explainable networks to map these embeddings with expression
- Provide larger transformer based decision layers for application oriented research if earlier endeavors are successful



Proposed architecture



Some pretrained candidates



Model	Average Pearson correlation
<u>H-Optimus-1</u>	0.423
<u>GenBio-PathFM</u>	0.42
<u>H-Optimus-0</u>	0.415
<u>UNI2-h</u>	0.414

Skin cutaneous melanoma is the category with highest correlation



First 3 Investigations

1. Explore the dataset

Examine spot counts, tissue/sample distribution, and gene-expression distributions.

2. Establish the baseline and Compare representations

Extract pretrained embeddings and train a simple regression model then compare selected pretrained encoders under the same evaluation setup.

3. Cross examining embedding values

Expected Compute Needs

- Laptop / Colab
- Possible GPU / cluster (16GB+)
- Need large storage



Questions

1. What are some tests suitable for embedding analysis?
2. Tips / Suggestions on handling large pretrained model?



Thank you

Project F: Spatial Gene Expression from Histology